Configurazione CLI Compagnia Theta

Il presente documento raccoglie i comandi CLI essenziali configurati su ciascun dispositivo dell'infrastruttura Compagnia Theta. Le configurazioni comuni a tutti gli switch di piano (VLAN access, trunk, native VLAN 99) sono descritte una sola volta con nota di applicabilità.

1. Switch di Piano Configurazione Comune

Applicata su tutti e 6 gli switch (Sw0_Piano_Terra → Sw5_Piano_Quinto). Cambia solo il numero VLAN (10/20/30/40/50/60) e l'hostname.

■ Sw0_Piano_Terra (esempio rappresentativo)

Catalyst Switch Piano Terra / VLAN 10

Hostname e VLAN:

```
Switch(config)# hostnameSw0_Piano_Terra
Switch(config)# vlan 10
Switch(config-vlan)# name VLAN0010
Switch(config-vlan)# exit
```

Porte access verso i PC (ripetuto per ogni porta occupata):

```
Sw0_Piano_Terra(config)# interface range fa0/1 - 20
Sw0_Piano_Terra(config-if-range)# switchport mode access
Sw0_Piano_Terra(config-if-range)# switchport access vlan 10
Sw0_Piano_Terra(config-if-range)# no shutdown
Sw0_Piano_Terra(config-if-range)# exit
```

Porta trunk verso Core Switch:

```
Sw0_Piano_Terra(config)# interface fa9/1
Sw0_Piano_Terra(config-if)# switchport mode trunk
Sw0_Piano_Terra(config-if)# switchport trunk native vlan 99
Sw0_Piano_Terra(config-if)# no shutdown
Sw0_Piano_Terra(config-if)# exit
Sw0_Piano_Terra# write memory
```

Nota: la stessa logica si applica a Sw1_Piano_Primo (VLAN 20), Sw2_Piano_Secondo (VLAN 30), Sw3_Piano_Terzo (VLAN 40), Sw4_Piano_Quarto (VLAN 50), Sw5_Piano_Quinto (VLAN 60).

2. Core Switch SW0_Core

■ SW0_Core

Catalyst Switch Aggregazione / Backbone

Creazione VLAN:

```
SW0_Core(config)# vlan 10,20,30,40,50,60,100,200,300
SW0_Core(config-vlan)# exit
```

Porte trunk verso switch di piano (Fa0/1→ Fa5/1) e verso Router (Fa9/1):

```
SW0_Core(config)# interface range fa0/1 - fa5/1
SW0_Core(config-if-range)# switchport mode trunk
SW0_Core(config-if-range)# switchport trunk native vlan 99
SW0_Core(config-if-range)# no shutdown
SW0_Core(config-if-range)# exit

SW0_Core(config)# interface fa9/1
SW0_Core(config-if)# switchport mode trunk
SW0_Core(config-if)# switchport trunk native vlan 99
SW0_Core(config-if)# no shutdown
SW0_Core(config-if)# exit
```

Porta verso NAS (Fa7/1) access VLAN 10:

```
SW0_Core(config)# interface fa7/1
SW0_Core(config-if)# switchport mode access
SW0_Core(config-if)# switchport access vlan 10
SW0_Core(config-if)# no shutdown
SW0_Core(config-if)# exit
SW0_Core# write memory
```

3. Router Centrale Router3

■ Router3 — Cisco Router Inter-VLAN Routing / DHCP / Default Route

Interfaccia uplink verso ASA (Fa8/0):

```
Router(config)# interface fa8/0
Router(config-if)# ip address 10.1.100.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
```

Subinterface inter-VLAN su Fa9/0 (schema comune ripetuto per ogni VLAN):

```
Router(config)# interface fa9/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 10.1.10.1 255.255.255.0
Router(config-subif)# ip helper-address 10.1.100.2
Router(config-subif)# exit

!Ripetere per VLAN 20 →60 con IP 10.1.X0.1 e encap dot1Q X0
```

DHCP Pool per ogni VLAN (schema comune):

```
Router(config)# ip dhcp pool VLAN10
Router(dhcp-config)# network 10.1.10.0 255.255.255.0
Router(dhcp-config)# default-router 10.1.10.1
Router(dhcp-config)# exit

!Ripetere per VLAN 20 →60 con network/default-router corrispondenti
```

Default route verso ASA:

```
Router(config)# ip route 0.0.0.0 0.0.0.0 10.1.100.2
Router# write memory
```

4. Firewall Perimetrale ASA 5505

■ **FW0_Piano_Terra** — Cisco ASA 5505 Firewall / NAT / ACL / DMZ

Assegnazione porte fisiche alle VLAN:

```
ciscoasa(config)# interface Ethernet0/0
ciscoasa(config-if)# switchport access vlan 1 ! → inside (verso Router)
ciscoasa(config-if)# no shutdown
ciscoasa(config-if)# exit

ciscoasa(config)# interface Ethernet0/2
ciscoasa(config-if)# switchport access vlan 3 ! → dmz (verso Server DMZ)
ciscoasa(config-if)# exit
```

Configurazione interfacce logiche:

```
ciscoasa(config)# interface Vlan1
ciscoasa(config-if)# nameif inside
ciscoasa(config-if)# security-level 100
ciscoasa(config-if)# ip address 10.1.100.2 255.255.255.0
ciscoasa(config-if)# exit

ciscoasa(config)# interface Vlan3
ciscoasa(config-if)# nameif dmz
ciscoasa(config-if)# security-level 50
ciscoasa(config-if)# ip address 10.1.200.1 255.255.255.0
ciscoasa(config-if)# forward interface Vlan1
ciscoasa(config-if)# exit
```

Nota: Vlan2 (outside, 200.0.0.1) è presente in configurazione ma senza nameif attivo per limite licenza base ASA 5505 in PT (max 2 interfacce con forward simultaneo). In produzione con licenza Security Plus tutte e tre le interfacce sarebbero attive.

NAT/PAT dinamico (overload):

```
ciscoasa(config)# object network INSIDE_NET
ciscoasa(config-network-object)# subnet 10.1.0.0 255.255.0.0
ciscoasa(config-network-object)# nat (inside,outside) dynamic interface
ciscoasa(config-network-object)# exit
```

Access Control List:

```
! ACL traffico inside → outside e inside → DMZ
ciscoasa(config)# access-list INSIDE_OUT permit ip any any
ciscoasa(config)# access-list INSIDE_OUT permit ip 10.1.0.0 255.255.0.0 any
ciscoasa(config)# access-list INSIDE_OUT permit ip 10.1.0.0 255.255.0.0 10.1.200.0
255.255.255.0
ciscoasa(config)# access-group INSIDE_OUT in interface inside

!ACL traffico DMZ → inside (blocco) e DMZ → outside (permesso)
ciscoasa(config)# access-list DMZ_IN permit icmp any any
ciscoasa(config)# access-list DMZ_IN deny ip any 10.1.0.0 255.255.0.0
ciscoasa(config)# access-list DMZ_IN permit ip any any
ciscoasa(config)# access-group DMZ_IN in interface dmz
```

Route statica verso reti interne e salvataggio:

```
ciscoasa(config)# route inside 10.1.0.0 255.255.0.0 10.1.100.1
ciscoasa# write memory
```

5. Server DMZ

■ Server_DMZ	Server PT Web Server in zona demilitarizzata

Configurazione IP (Desktop → IP Configuration):

```
IP Address : 10.1.200.2
Subnet Mask : 255.255.255.0
Gateway : 10.1.200.1 (ASA DMZ interface)
```

Servizi attivi (Desktop → Services):

```
HTTP : ON (accesso web simulato)
HTTPS : ON
```

6. NAS (Network Attached Storage)

■ NAS_Piano_Terra	Server PT Storage condiviso / VLAN 10

Configurazione IP (Desktop → IP Configuration):

```
IP Address : 10.1.10.5
Subnet Mask : 255.255.255.0
Gateway : 10.1.10.1 (Router subinterface VLAN 10)
```

Servizi attivi:

```
HTTP : ON (simulazione interfaccia web NAS)
FTP : ON (simulazione trasferimento file)
```

Il NAS è collegato direttamente al Core Switch (porta Fa7/1, access VLAN 10) ed è raggiungibile da tutti i piani tramite routing inter-VLAN.